

Lab 11: Priors as Fake Data

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11.1 NBA Free Throws Revisited

Labs 8 and 9 treated each player's true free-throw probability p_i as fixed and used confidence intervals and the bootstrap to describe sampling uncertainty. Today you will use a Beta prior:

$$\begin{aligned} W_i \mid p_i &\sim \text{Binomial}(N_i, p_i), \\ p_i &\sim \text{Beta}(\alpha, \beta), \\ p_i \mid W_i, L_i &\sim \text{Beta}(\alpha + W_i, \beta + L_i). \end{aligned}$$

The main summaries are

$$\begin{aligned} \hat{p}_i^{\text{mean}} &= \frac{W_i + \alpha}{N_i + \alpha + \beta}, \\ \hat{p}_i^{\text{MAP}} &= \frac{W_i + \alpha - 1}{N_i + \alpha + \beta - 2}, \end{aligned}$$

when the posterior mode is interior.

11.1.1 Task 1: Prepare the Data

Using `11_nba-free-throws.csv`, create one row per player and calculate approximate season totals:

$$\begin{aligned} W_i &= \text{round}(G_i \cdot FT_i), \\ N_i &= \text{round}(G_i \cdot FTA_i), \\ L_i &= N_i - W_i. \end{aligned}$$

When a player appears for multiple teams, retain the TOT row.

Select at least six players spanning low, medium, and high attempt totals.

11.1.2 Task 2: Compare Three Priors

For each selected player, compute posterior means, MAP estimates, and 95% credible intervals under:

1. a weak prior, $\text{Beta}(2, 2)$;
2. a league-centered prior whose center is the overall free-throw percentage and whose strength is 30 attempts;
3. a prior of your design for an elite free-throw shooter.

For the designed prior, state the substantive projection used for its center and how many attempts of evidence its strength represents. Plot the MLE and posterior means with their credible intervals. Explain which players move most and why.

11.1.3 Task 3: Prior Sensitivity

Choose one low-attempt player and one high-attempt player. Vary prior strength over

$$\{2, 10, 30, 100, 300\}$$

while holding the prior center fixed at the league rate. Plot posterior mean against prior strength for both players. At what point does the prior dominate the data?

Then split one player's attempts into two batches. Verify that using the first posterior as the second prior gives the same final posterior as updating with all attempts at once.

11.1.4 Task 4: Posterior Prediction

For each selected player, simulate 5,000 possible outcomes for their next 50 attempts:

1. draw $p_i^{(m)}$ from the player's posterior;
2. draw $W_{i,\text{future}}^{(m)} \sim \text{Binomial}(50, p_i^{(m)})$.

Convert each draw to a future free-throw percentage $W_{i,\text{future}}^{(m)}/50$. Report its predictive mean and a 95% posterior predictive interval. Compare its width with the credible interval for p_i and explain why they differ.

Confirm that the simulated predictive mean agrees with the Beta-Binomial expectation

$$50 \frac{\alpha + W_i}{\alpha + \beta + N_i}.$$

11.1.5 Task 5: Model Critique

State two assumptions of the Beta-Binomial model that are questionable for NBA free throws. For each, describe one model extension or data feature that could address it.